

SIEM-Integrated Network Intrusion Detection System (NIDS) - SOC Platform

SOC PLATFORM - CYBERSECURITY LAB ASSESSMENT REPORT

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Project Title	SIEM-Integrated Network Intrusion Detection System (NIDS) - SOC Platform
Classification	Confidential Lab Use Only
Assessment type	Network Intrusion detection + SIEM Log analysis
Platform	Python/Scapy IDS, Splunk Enterprise SIEM
Lab Environment	Virtual Box : Kali linux(2) , Metasploitable3
Tools used	Python 3, Scapy, Splunk, Nmap, Hydra, hping3, arpspoof, Wireshark
Detection rules	6 Rules : Port Scan, ARP Spoof, Brute Force, DNS Tunneling, ICMP Flood, SYN Flood
Date	24 May 2026
Status	Final

Executive Summary:

The report documents the design , implementation and working of a SIEM integrated Network Intrusion Detection System (NIDS) which is a full SOC platform built entirely in a controlled virtual lab environment. The project combines a custom Python/Scapy-based NIDS with Splunk Enterprise SIEM to simulate a real-world Security Operations Centre (SOC) workflow.

To detect cyber attacks, six rules were implemented. These rules covered a range of attacks like port scanning, ARP spoofing, SSH brute force, DNS tunneling, ICMP flooding, and SYN flood DoS attack. these attack were simulated from an attacker VM , detected by NIDS in real time and automatically forwarded to splunk for log aggregation, correlation & dashboard visualisation while the packets were captured for further analysis.

Threat	CVSS Score	Severity	Detected	In Splunk
Port Scan	7.5	High	Yes	Yes
ARP spoofing	8.1	High	Yes	Yes
SSH brute force	9.8	Critical	Yes	Yes
DNS tunneling	7.5	High	Yes	Yes
ICMP flood/ ping of death	7.5	High	Yes	Yes
SYN flood / DoS	8.6	High	Yes	Yes

Methodology:

Lab Environment:

All the simulation was done in an isolated Virtual Box host-only network with no internet connectivity. Three virtual machines were deployed :

- **Kali Linux (Attacker)** - 192.168.56.101 , used to simulate all the attack traffic using Nmap, Hydra and arpspoof.
- **Metasploitable3 (Victim)** - 192.168.56.102 , an intentionally chosen vulnerable linux VM used as the target for all the attacks.
- **Kali Linux (IDS Monitor)** - 192.168.56.103, another kali linux VM to use as IDS and network adapter configured in promiscuous mode to capture all the traffic. The Python/ Scapy IDS ran on this VM.

Platform architecture:

The Soc platform contain two integrated components:

- Custom Python/Scapy NIDS: Monitors live network traffic in promiscuous mode, applies 6 detection rules using sliding-window packet analysis, and generates structured alert logs (CSV/JSON) in real time.
- Splunk Enterprise SIEM: Monitors the IDS alert log file via a file input data source, ingests alerts automatically, and presents them on a custom SOC dashboard with panels for alert counts, severity breakdown, attack timeline, and alert log table.

Methodology Phase:

The following steps were followed to structured and execute the project:

- **Step I** : Environment setup - Virtual lab configuration to isolate the network setting the adapter with host-only network. Only the Kali monitor was set to promiscuous activation.
The dependencies on the monitor VM were pre installed. The verification of installation was done before simulating the attacks.
- **Step II** : IDS development - A custom python/ scapy script was written with sliding window detection rules for each categorised threat. While using it the default interface was set to perform. `sudo python ids_monitor2.py`
Customization of detection threshold was scripted for automation previously in `ids_monitor2.py` in following way
`PORT_SCAN_THRESHOLD=50`
`PORT_SCAN_WINDOW_SEC=5`
`BRUTE_FORCE_THRESHOLD=10`
`BRUTE_FORCE_WINDOW_SEC=60`
`BRUTE_FORCE_PORTS={22,21,23,3389,5900}`

DNS_QUERY_THRESHOLD = 10

DNS_WINDOW_SEC = 30

ICMP_THRESHOLD = 100

ICMP_WINDOW_SEC = 5

SYN_FLOOD_THRESHOLD = 500

SYN_FLOOD_WINDOW_SEC = 3

Detection rules that were implemented are

1. Port scan detection : tracks SYN packets per source IP using a sliding window. Fires when count exceeds the threshold.
 2. Brute force detection : monitors SYN packets to service ports like SSH,FTP,Telnet, RDP, VNC per source IP.
 3. ARP spoofing detection : configured a trusted IP in MAC table. It fires when a known IP claims a different MAC address.
 4. DNS tunnelling: implemented detection rule to identify high volume query pattern from attacker's machine
 5. ICMP flood detection: configured type8 based ICMP filtering protocol aware approach that detects attack accurately across any network topology.
 6. SYN flood detection: implemented a detection logic to differentiate reconnaissance activity from volumetric DoS attacks based on destination port distribution pattern.
- **Step III:** SIEM integration- the splunk enterprise was installed on monitor VM, file input configured to ingest IDS alert CSV and a custom SOC dashboard was built with visualisation panel to detect the threat better.
 - **Step IV:** Attack simulation - controlled attacks were launched from the attacker VM to validate each detection rule and monitored it from IDS .
For port scan : nmap -sS -T4 192.168.56.102
For brute force : hydra -l root -P /usr/share/wordlists/rockyou.txt -t 4 ssh://192.168.56.102
(if required wordlists should be extracted using gunzip)
For ARP spoof : sudo arpspoof -i eth1 -t 192.168.56.102 192.168.56.103
For DNS tunnelling: for i in {1..60}; do dig @192.168.56.103 test.local; done
For ICMP flood detection: sudo hping3 -1 -flood 192.168.56.102
For SYN flood detection : sudo hping3 -S -flood -p 80 192.168.56.102
 - **Step V:** Forensic Evidence - From IDS monitor vm using tshark, the network packets were captured and filtered using Wireshark for further analysis of the identified/detected attack, saved as ids_filtered.pcap .
 - **Step VI:** Analysis and Report - Alert logs were reviewed through ids_alert.csv and ids_alert.json file. Then CVSS scores were assigned to each detected based on their severity to threat and remediations were recommended in documentation to prevent any further compromise of security in the network.

SIEM Integration:

Architecture:

Splunk Enterprise 8.2.6 was deployed on the IDS Monitor VM (192.168.56.103). An input data file as source was configured to monitor the IDS alert log file (ids_alerts.csv) in real time. Therefore, every alert generated by the python NIDS was automatically ingested into Splunk's ids_alerts index within seconds of detection.

Panel	SPL query	Purpose
Total Alerts	index=ids_alerts stats count	Overall alert volume
Critical Alerts	index=ids_alerts severity=CRITICAL stats count	Critical alerts only
High Alerts	index=ids_alerts severity=HIGH stats count	High alerts volume
Alert Timeline	index=ids_alerts timechart count by threat_type	Attack timeline chat evolution
Alert Severity	index=ids_alerts stats count by severity	Severity pie chart
Threat Detected	index=ids_alerts stats count by threat_type sort -count	Column chart for detected threat
Alert Log	index=ids_alerts table timestamp src_ip dst_ip threat_type severity	Full alert log

Observations:

- 21 alerts were successfully ingested into Splunk across 6 distinct threat categories within seconds of NIDS detection demonstrating efficient real time SIEM integration where total alerts are 21 among them 2 are critical and rest 19 are high on severity.
- The Alerts timeline chat shows the captured timeline between all the attacks phase like Port scan to SYN flood/DoS attack that can be distinguished easily , occurring between 02:42–02:47 on May 24 2026 ,
- The severity pie chart is efficient for demonstrating the high and critical attacks which makes it easy to decide on prioritising the remediation accordingly.
- Splunk's search capabilities allowed retrospective analysis of all alerts, enabling correlation of attack sequences that would be difficult to identify from raw logs alone.

Project Evolution & Engineering Decisions:

This project underwent several significant design iterations during the implementation stage. Each occurred challenge was diagnosed systematically and resolved, reflecting real world SOC engineering practice where solutions must be adapted to infrastructure constraints.

Multi Interface packet capture: initial deployment on single interface eth0 resulted in incomplete packet capture therefore traffic between VMs flowed across both eth0 and eth1 due to VirtualBox's virtual adapter assignment behaviour. So, Scapy was reconfigured to sniff on both interfaces simultaneously (IFACE=["eth0", "eth1"]), restoring the complete network visibility.

Port Scan vs SYN flood differentiation: Both port scanning and SYN flood attacks use TCP SYN packets which causes the original single detection rule to trigger duplicate alerts at the same time. The rules were separated by adding a unique destination port count check. Port scan detection looks for SYNs targeting multiple ports (`unique_ports > 1`). In contrast, SYN flood detection activates when there are high volumes of SYNs to a single port (500 or more packets in 3 seconds).

ICMP False positive elimination: The ICMP flood detection rule was initially configured for bidirectional alerts on every incident. One for the attacker's echo requests and another one, for the victim's automatic echo replies, Since the monitor was operated on promisc mode capturing all the traffic. This problem was resolved by filtering the ICMP type 8 packets, which are the echo request packets. This way of doing things works with any kind of network without requiring specific IP addresses.

DNS tunnelling detection : it required 3 changes or fixation, first DNS QR flag filter (`qr=0`) was added to capture only query packets and exclude the monitor vm's own DNS responses; second , threshold comparison was changed to equality (`==`) to greater than equal (`>=`) to prevent the counter skipping of threshold value under high packets rate and lastly , a per source IP alert cooldown set was configured to prevent repetition of alerts from the same attacker after threshold was reached.

SIEM Integration: the original project was limited to standalone NIDS with CSV/JSON alert logs. Splunk was integrated to replicate a production SOC workflow where IDS alerts forwarded for log aggregation, correlation, and dash visualisation to increase the project's alignment with real world SOC operation.

Capture Method: when running the IDS script and Splunk simultaneously with a live Wireshark GUI capture on a 3GB VM, there were constant system freezes and OOM process kills. So, the capture strategy was adapted to use tshark CLI for real-time packet capture, a lightweight alternative with much lower memory footprint. Wireshark was used exclusively for post-capture forensic analysis of the saved pcap file.

Findings:

1. Port Scan detection:

A TCP SYN stealth scan was simulated from the attacker VM targeting the Metasploitable3. This technique sends SYN packets to multiple ports without completing the handshake, allowing the attacker to enumerate an open service while the basic connection logs were evaded. The IDS monitor kept track of SYN packets counts per source IP using a 5 second sliding window and fired high severity alerts upon reaching the threshold. Then the packets were captured using wireshark by filtering out `tcp.flags.syn == 1 && tcp.flags.ack == 0` for live network traffic analysis.

- **Threat type: Port Scanning(SYN stealth)**
- **Tools used: nmap -sS -T4**
- **CVSS score: 7.5 (high)**
- **Detection rule: 50+ SYN packets from same source IP**
- **Alert Triggered: Yes**
- **Splunk Ingested: Yes, appeared in dashboard immediately**

2. Brute Force Login Detection:

A dictionary based brute force attack was done against SSH on port 22 of targeted Metasploitable3 VM using Hydra with rockyou.txt password list containing 14 million common passwords. The IDS monitored SYN connection attempted per source IP to a familiar service port within 60 seconds of sliding window. The High volume of consecutive attempts were rapidly simulated from source IP that triggered an immediate high alert which is highest in this assessment. Those high alerted live packets were captured with wireshark as well by filtering with `tcp.dport==22`.

- **Threat type: Brute Force credential attack**
- **Tools used: hydra with rockyou.txt wordlist**
- **CVSS score: 9.8 Critical**
- **Detection rule: 10+ SYN packets from same source IP to service port within 60s**
- **Alert Triggered: Yes, alert fired within seconds of hydra execution.**
- **Splunk Ingested: Yes, displayed in critical panels**

3. ARP Spoofing / MITM Detection:

To the victim VM the attacker used arpspoof to send forged ARP replies, poisoning its ARP cache data and redirected the traffic. The IDS maintained a trusted IP to MAC mapping table built from ARP replies. The replies that claimed a different MAC address for known IP, flagged as ARP spoofing attempt with conflicted MAC

address. The live packets were captured with Wireshark by filtering out with **arp**. (note: as the controlled lab environment was set to Host only network, it lacked a physical gateway at 192.168.56.1 in that case the monitor VM IP was used to impersonate as target IP. This doesn't affect the validity of the detection mechanism).

- **Threat type: ARP spoofing / Man in the Middle(MITM)**
- **Tools used: arpspoof(dsniff package)**
- **CVSS score: 8.1 Critical**
- **Detection rule: Mac address change detected for a previously trusted IP address.**
- **Alert Triggered: Yes**
- **Splunk Ingested: Yes, MAC conflict record shows**

4. DNS Tunneling Detection:

DNS tunneling is a covert data exfiltration technique that encodes the data inside DNS query names to bypass firewalls. High volume DNS query patterns were generated from the attacker VM to simulate the attack which IDS detected successfully identifying the anomalous query volume using a 10-second sliding window and fired a HIGH severity alert. Due to lab environment constraints (bind9 service instability on host-only network), bidirectional DNS traffic was captured, both attacker queries and monitor responses were flagged, demonstrating promiscuous mode capture effectiveness. Those high alert live packets were captured with Wireshark as well by filtering with **dns**.

- **Threat type: DNS tunneling / data exfiltration via DNS**
- **Tools used: for i in {1..60}; do dig @192.168.56.103 test.local; done**
- **CVSS score: 7.5 High**
- **Detection rule: 50+ DNS queries from same IP within 10-second window OR query name >50 chars**
- **Alert Triggered: Yes, alert fired after 50 queries**
- **Splunk Ingested: Yes, visible in alerts by threat panel**

5. ICMP Flood / Ping of Death Detection:

An ICMP flood attack was generated from the attacker VM using tool hping3 to overwhelm the victim with ping packets consuming the bandwidth and processing resources. The IDS detected a huge flood which is high in volume, using a 5-second sliding window and fired a HIGH severity alert. To filter those flooded high alert live packets were captured with Wireshark filter **icmp**.

- **Threat type: ICMP flood / ping of death, volumetric DoS**
- **Tools used: hping3**

- **CVSS score: 7.5 High**
- **Detection rule: 100+ ICMP packets were captured from same IP within 5 seconds**
- **Alert Triggered: Yes, alert fired during flood**
- **Splunk Ingested: Yes, visible in alerts by threat panel**

6. SYN Flood / DoS Detection:

A SYN flood attack targeted port 80 or HTTP on Metasploitable3 sending massive SYN packets without completing the TCP handshake to exhaust the server's half-open connection table. Unlike port scanning which targets many ports, this attack targeted one port with overwhelming volume. The IDS correctly distinguished this from port scan behaviour and fired a separate SYN flood alert.

- **Threat type: SYN flood / TCP Denial of Service**
- **Tools used: hping3 (-S -flood -p 80)**
- **CVSS score: 8.6 High**
- **Detection rule: 200+ SYN packets were captured from same IP within 5 seconds**
- **Alert Triggered: Yes, alert fired within seconds**
- **Splunk Ingested: Yes, recorded in Alert log table**

Risk Summary :

Findings	CVSS Score	Severity	Detected	Priority
Brute Force login attempt	9.8	critical	yes	p1
SYN flood/DoS	8.6	high	yes	p2
ARP Spoofing	8.1	high	yes	p3
Port Scan	7.5	high	yes	p4
DNS Tunneling	7.5	high	yes	p5
ICMP flood	7.5	high	yes	p6

Note: The total number of alerts was 21 with 2 CRITICAL and 19 HIGH severity alerts as the SYN flood attack generated the highest number of alerts due to its volumetric nature.

Evidence:

The following part contains the evidence that is collected during the assessment to analyse the entire process and remediate the situation. Alert logs were automatically created by the IDS (as it was scripted) and saved to ids_alerts.csv and ids_alerts.json on the kali linux monitor.

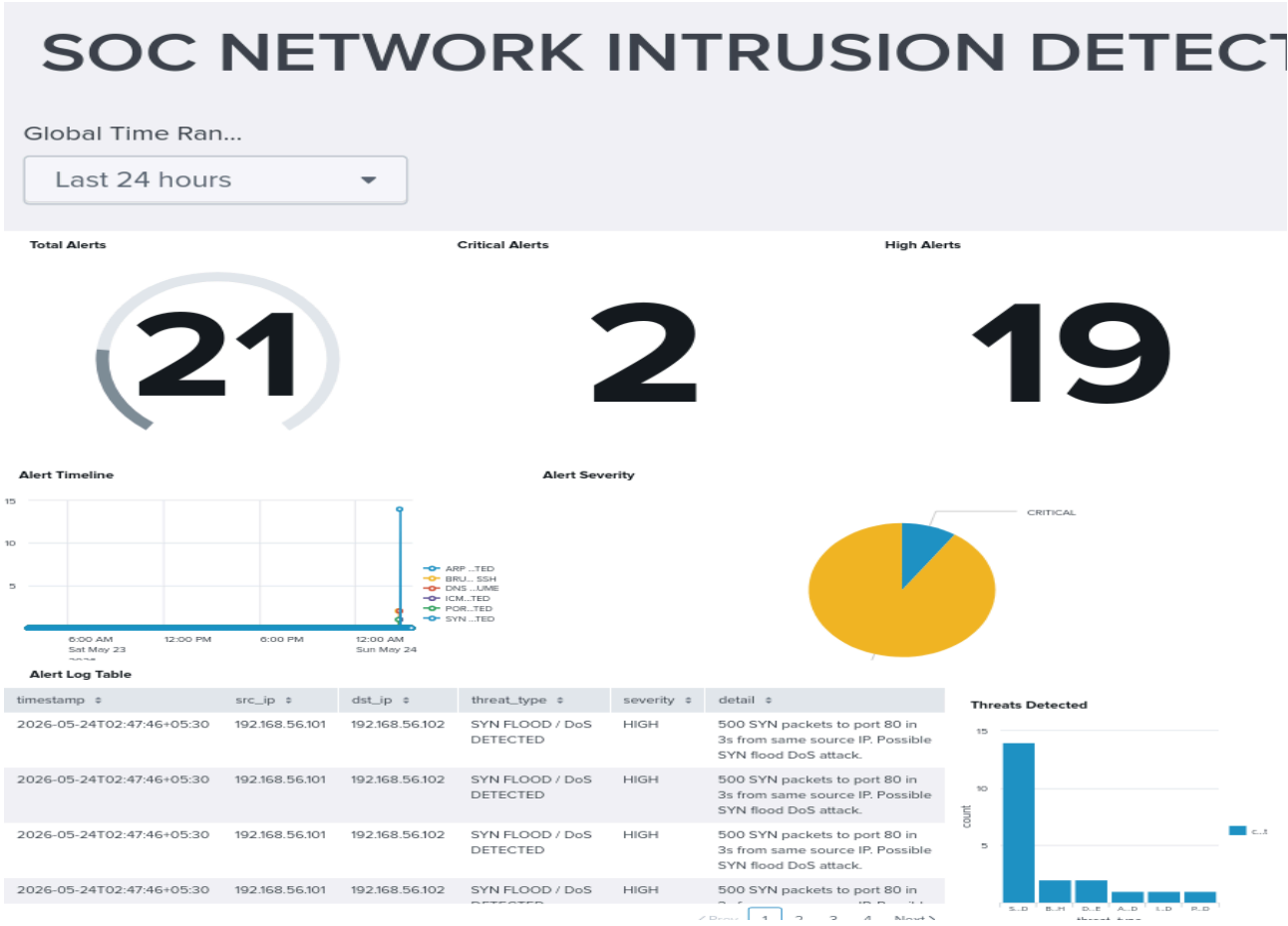
Alert log format: Each alert entry contains the .json file following field:-

Timestamp= for date and time; src_ip= Source IP; dst_ip=Destination IP;

threat_type=classification of the detected threat; severity=rating; detail= readable desc;

Log Snippet:

```
"timestamp": "2026-05-24 02:43:31",
"src_ip": "192.168.56.101",
"dst_ip": "192.168.56.102",
"threat_type": "BRUTE FORCE DETECTED \u2014 SSH",
"severity": "CRITICAL",
"detail": "10 connection attempts to SSH (port 22) in 60s."
},
{
"timestamp": "2026-05-24 02:45:34",
"src_ip": "08:00:27:c5:06:86",
"dst_ip": "192.168.56.103",
"threat_type": "ARP SPOOFING DETECTED",
"severity": "HIGH",
"detail": "IP 192.168.56.103 previously mapped to MAC 08:00:27:44:dd:84, now claiming MAC 08:00:27:c5:06:86. Possible MITM attack."
},
{
"timestamp": "2026-05-24 02:45:53",
"src_ip": "192.168.56.101",
"dst_ip": "192.168.56.103",
"threat_type": "DNS TUNNELING DETECTED \u2014 HIGH QUERY VOLUME",
"severity": "HIGH",
"detail": "10 DNS queries in 30s from same IP. 1 unique domains queried.Possible data exfiltration via DNS."
},
}
```



Splunk dashboard screenshot

```
(md@kali)-[~]
$ sudo python3 ids_monitor2.py

Network IDS — Python/Scapy

Interface : ['eth0', 'eth1']
Log (CSV) : ids_alerts.csv
Log (JSON): ids_alerts.json
Thresholds:
  Port scan      → 50 SYNs in 5s
  Brute force    → 10 SYNs to service port in 60s
  ARP spoof      → any MAC change for known IP
  DNS Tunneling  → (CVSS 7.5 HIGH)
  ICMP Flood / Ping of Death → (CVSS 7.5 HIGH)
  SYN Flood / DoS Detection → (CVSS 8.6 HIGH)

[*] Sniffing... Press Ctrl+C to stop.

[2026-05-24 02:42:22] [HIGH] PORT SCAN DETECTED
SRC: 192.168.56.101 → DST: 192.168.56.102
50 SYN packets in 5s — 36 unique destination ports targeted.

[2026-05-24 02:43:19] [CRITICAL] BRUTE FORCE DETECTED — SSH
SRC: 192.168.56.101 → DST: 192.168.56.102
10 connection attempts to SSH (port 22) in 60s.

[2026-05-24 02:43:31] [CRITICAL] BRUTE FORCE DETECTED — SSH
SRC: 192.168.56.101 → DST: 192.168.56.102
10 connection attempts to SSH (port 22) in 60s.

[2026-05-24 02:45:34] [HIGH] ARP SPOOFING DETECTED
SRC: 08:00:27:c5:06:86 → DST: 192.168.56.103
IP 192.168.56.103 previously mapped to MAC 08:00:27:44:dd:84, now claiming MAC 08:00:27:c5:06:86. Possible MITM attack.

[2026-05-24 02:45:53] [HIGH] DNS TUNNELING DETECTED — HIGH QUERY VOLUME
SRC: 192.168.56.101 → DST: 192.168.56.103
10 DNS queries in 30s from same IP. 1 unique domains queried. Possible data exfiltration via DNS.

[2026-05-24 02:45:53] [HIGH] DNS TUNNELING DETECTED — HIGH QUERY VOLUME
SRC: 192.168.56.103 → DST: 192.168.56.101
10 DNS queries in 30s from same IP. 1 unique domains queried. Possible data exfiltration via DNS.

[2026-05-24 02:46:44] [HIGH] ICMP FLOOD DETECTED
SRC: 192.168.56.101 → DST: 192.168.56.102
100 ICMP packets in 5s from same IP. Possible DoS/DDoS attack via ICMP flood.
```

IDS alerts

```
10 DNS queries in 30s from same IP. 1 unique domains queried. Possible data exfiltration via DNS.

[2026-05-24 02:46:44] [HIGH] ICMP FLOOD DETECTED
SRC: 192.168.56.101 → DST: 192.168.56.102
100 ICMP packets in 5s from same IP. Possible DoS/DDoS attack via ICMP flood.

[2026-05-24 02:47:13] [HIGH] SYN FLOOD / DoS DETECTED
SRC: 192.168.56.101 → DST: 192.168.56.102
500 SYN packets to port 80 in 3s from same source IP. Possible SYN flood DoS attack.

[2026-05-24 02:47:45] [HIGH] SYN FLOOD / DoS DETECTED
SRC: 192.168.56.101 → DST: 192.168.56.102
500 SYN packets to port 80 in 3s from same source IP. Possible SYN flood DoS attack.

[2026-05-24 02:47:45] [HIGH] SYN FLOOD / DoS DETECTED
SRC: 192.168.56.101 → DST: 192.168.56.102
500 SYN packets to port 80 in 3s from same source IP. Possible SYN flood DoS attack.

[2026-05-24 02:47:46] [HIGH] SYN FLOOD / DoS DETECTED
SRC: 192.168.56.101 → DST: 192.168.56.102
500 SYN packets to port 80 in 3s from same source IP. Possible SYN flood DoS attack.

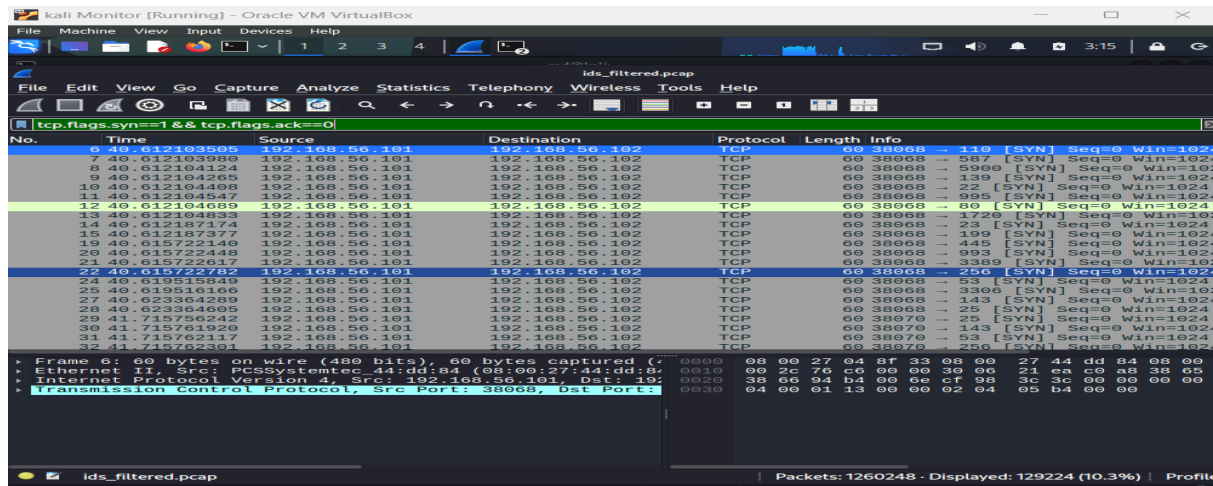
[2026-05-24 02:47:46] [HIGH] SYN FLOOD / DoS DETECTED
SRC: 192.168.56.101 → DST: 192.168.56.102
500 SYN packets to port 80 in 3s from same source IP. Possible SYN flood DoS attack.

[2026-05-24 02:47:46] [HIGH] SYN FLOOD / DoS DETECTED
SRC: 192.168.56.101 → DST: 192.168.56.102
500 SYN packets to port 80 in 3s from same source IP. Possible SYN flood DoS attack.

[2026-05-24 02:47:46] [HIGH] SYN FLOOD / DoS DETECTED
SRC: 192.168.56.101 → DST: 192.168.56.102
500 SYN packets to port 80 in 3s from same source IP. Possible SYN flood DoS attack.

[2026-05-24 02:47:46] [HIGH] SYN FLOOD / DoS DETECTED
SRC: 192.168.56.101 → DST: 192.168.56.102
500 SYN packets to port 80 in 3s from same source IP. Possible SYN flood DoS attack.
```

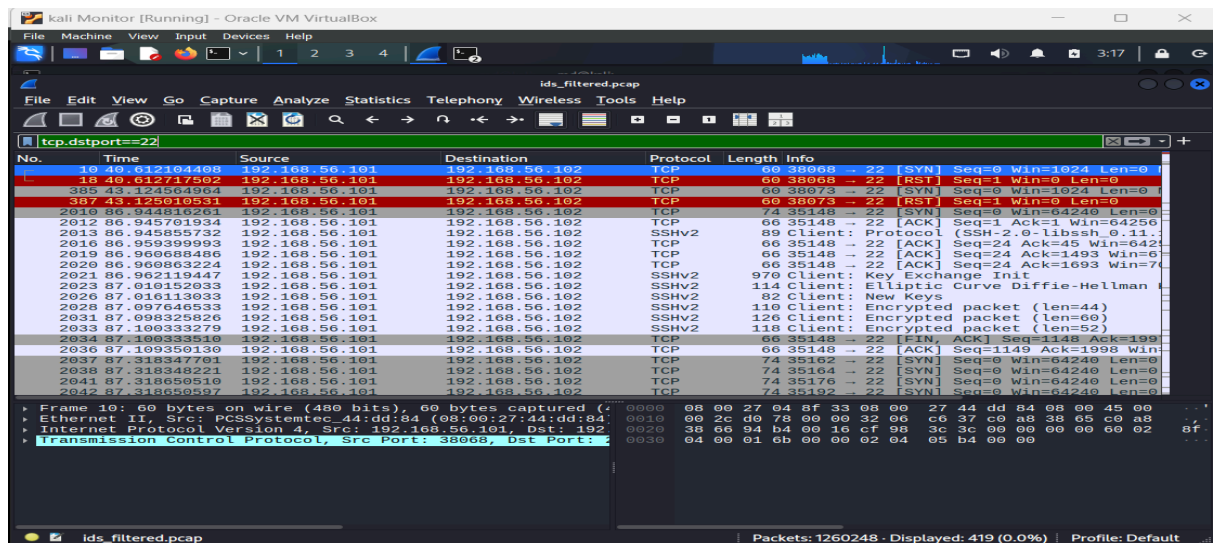
IDS alerts



The image shows a Wireshark capture of port scan packets. The filter is set to `tcp.flags.syn==1 && tcp.flags.ack==0`. The capture shows multiple SYN packets from 192.168.56.101 to 192.168.56.102. The packet details pane shows the Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol layers.

No.	Time	Source	Destination	Protocol	Length	Info
6	40.612103665	192.168.56.101	192.168.56.102	TCP	60	38068 → 116 [SYN] Seq=0 Win=102
7	40.612103990	192.168.56.101	192.168.56.102	TCP	60	38068 → 557 [SYN] Seq=0 Win=102
8	40.612104124	192.168.56.101	192.168.56.102	TCP	60	38068 → 5900 [SYN] Seq=0 Win=102
9	40.612104265	192.168.56.101	192.168.56.102	TCP	60	38068 → 139 [SYN] Seq=0 Win=102
10	40.612104408	192.168.56.101	192.168.56.102	TCP	60	38068 → 22 [SYN] Seq=0 Win=102
11	40.612104547	192.168.56.101	192.168.56.102	TCP	60	38068 → 995 [SYN] Seq=0 Win=102
12	40.612104689	192.168.56.101	192.168.56.102	TCP	60	38068 → 80 [SYN] Seq=0 Win=102
13	40.612104833	192.168.56.101	192.168.56.102	TCP	60	38068 → 1720 [SYN] Seq=0 Win=102
14	40.612104974	192.168.56.101	192.168.56.102	TCP	60	38068 → 23 [SYN] Seq=0 Win=102
15	40.612105117	192.168.56.101	192.168.56.102	TCP	60	38068 → 199 [SYN] Seq=0 Win=102
16	40.612105260	192.168.56.101	192.168.56.102	TCP	60	38068 → 445 [SYN] Seq=0 Win=102
20	40.615722448	192.168.56.101	192.168.56.102	TCP	60	38068 → 993 [SYN] Seq=0 Win=102
21	40.615722617	192.168.56.101	192.168.56.102	TCP	60	38068 → 3389 [SYN] Seq=0 Win=102
22	40.615722782	192.168.56.101	192.168.56.102	TCP	60	38068 → 240 [SYN] Seq=0 Win=102
24	40.619515849	192.168.56.101	192.168.56.102	TCP	60	38068 → 53 [SYN] Seq=0 Win=102
25	40.619516006	192.168.56.101	192.168.56.102	TCP	60	38068 → 3305 [SYN] Seq=0 Win=102
27	40.623364289	192.168.56.101	192.168.56.102	TCP	60	38068 → 143 [SYN] Seq=0 Win=102
28	40.623364665	192.168.56.101	192.168.56.102	TCP	60	38068 → 25 [SYN] Seq=0 Win=102
39	41.715762412	192.168.56.101	192.168.56.102	TCP	60	38070 → 25 [SYN] Seq=0 Win=102
30	41.715762920	192.168.56.101	192.168.56.102	TCP	60	38070 → 143 [SYN] Seq=0 Win=102
31	41.715763117	192.168.56.101	192.168.56.102	TCP	60	38070 → 63 [SYN] Seq=0 Win=102
32	41.715763301	192.168.56.101	192.168.56.102	TCP	60	38070 → 250 [SYN] Seq=0 Win=102

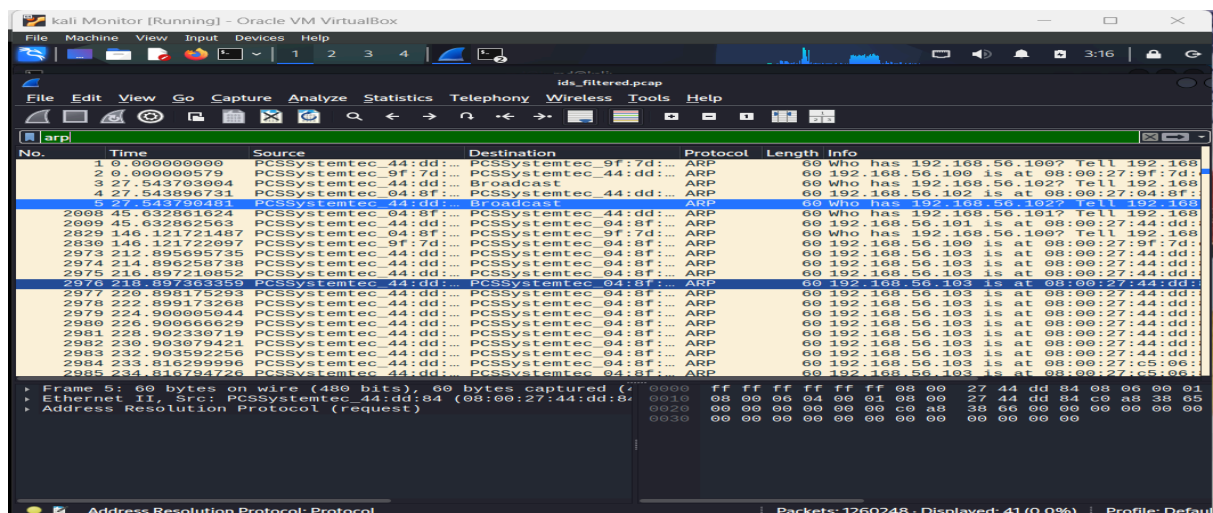
Port scan packets capture



The image shows a Wireshark capture of brute force login packets. The filter is set to `tcp.dstport==22`. The capture shows multiple SSH packets from 192.168.56.101 to 192.168.56.102. The packet details pane shows the Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol layers.

No.	Time	Source	Destination	Protocol	Length	Info
10	40.612104488	192.168.56.101	192.168.56.102	TCP	60	38068 → 22 [SYN] Seq=0 Win=1024 Len=0
18	40.612717502	192.168.56.101	192.168.56.102	TCP	60	38068 → 22 [RST] Seq=0 Win=0 Len=0
385	43.124504964	192.168.56.101	192.168.56.102	TCP	60	38073 → 22 [SYN] Seq=0 Win=1024 Len=0
387	43.125010531	192.168.56.101	192.168.56.102	TCP	60	38073 → 22 [RST] Seq=0 Win=0 Len=0
2010	86.944816261	192.168.56.101	192.168.56.102	TCP	74	35148 → 22 [SYN] Seq=0 Win=64240 Len=0
2012	86.945761934	192.168.56.101	192.168.56.102	TCP	66	35148 → 22 [ACK] Seq=1 Ack=1 Win=64250
2013	86.945855732	192.168.56.101	192.168.56.102	SSHv2	89	Client: Protocol (SSH-2.0-libssh_0.1)
2016	86.959399993	192.168.56.101	192.168.56.102	TCP	66	35148 → 22 [ACK] Seq=2 Ack=45 Win=642
2019	86.96068486	192.168.56.101	192.168.56.102	TCP	66	35148 → 22 [ACK] Seq=24 Ack=1493 Win=6
2020	86.960803224	192.168.56.101	192.168.56.102	TCP	66	35148 → 22 [ACK] Seq=24 Ack=1693 Win=7
2021	86.962119447	192.168.56.101	192.168.56.102	SSHv2	970	Client: Key Exchange Init
2023	87.008325926	192.168.56.101	192.168.56.102	SSHv2	114	Client: Elliptic Curve Diffie-Hellman
2026	87.016113933	192.168.56.101	192.168.56.102	SSHv2	82	Client: New Keys
2028	87.097646533	192.168.56.101	192.168.56.102	SSHv2	110	Client: Encrypted packet (len=44)
2031	87.098325926	192.168.56.101	192.168.56.102	SSHv2	126	Client: Encrypted packet (len=60)
2033	87.100333279	192.168.56.101	192.168.56.102	SSHv2	118	Client: Encrypted packet (len=52)
2034	87.100333510	192.168.56.101	192.168.56.102	TCP	66	35148 → 22 [FIN, ACK] Seq=1148 Ack=199
2036	87.100333510	192.168.56.101	192.168.56.102	TCP	66	35148 → 22 [ACK] Seq=1149 Ack=1998 Win=
2037	87.318347791	192.168.56.101	192.168.56.102	TCP	74	35162 → 22 [SYN] Seq=0 Win=64240 Len=0
2038	87.318348221	192.168.56.101	192.168.56.102	TCP	74	35164 → 22 [SYN] Seq=0 Win=64240 Len=0
2041	87.318656510	192.168.56.101	192.168.56.102	TCP	74	35176 → 22 [SYN] Seq=0 Win=64240 Len=0
2042	87.318656597	192.168.56.101	192.168.56.102	TCP	74	35192 → 22 [SYN] Seq=0 Win=64240 Len=0

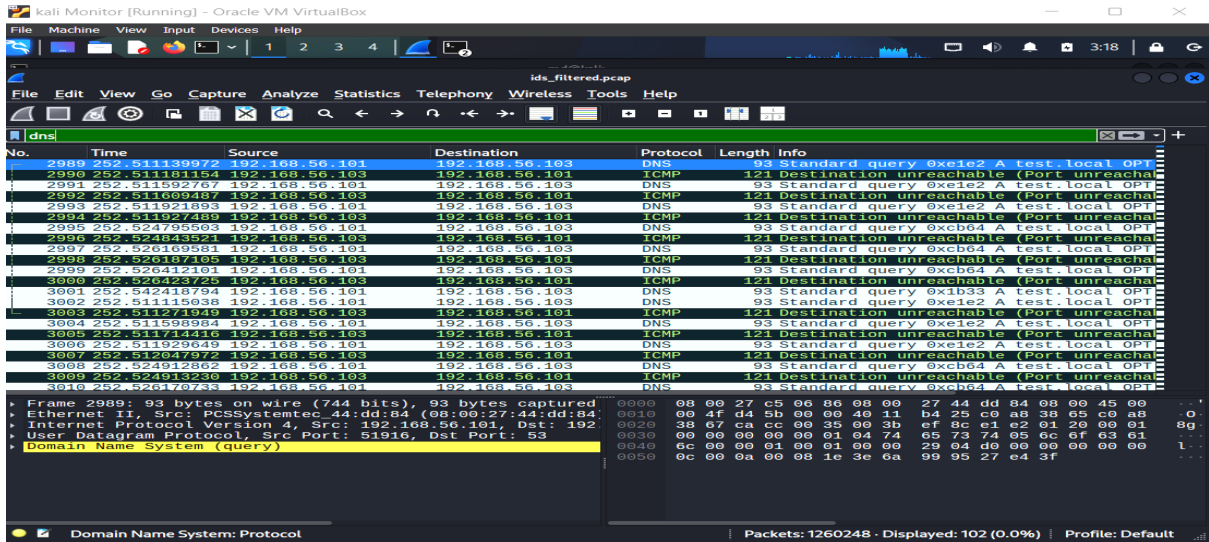
Brute force login packets capture



The image shows a Wireshark capture of ARP spoofing packets. The filter is set to `arp`. The capture shows multiple ARP requests from 192.168.56.101 to 192.168.56.102. The packet details pane shows the Ethernet II, Internet Protocol Version 4, and Address Resolution Protocol layers.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	PCSSystemtec_44:dd:...	PCSSystemtec_9f:7d:...	ARP	60	Who has 192.168.56.100? Tell 192.168.
2	0.000000059	PCSSystemtec_44:dd:...	Broadcast	ARP	60	Who has 192.168.56.102? Tell 192.168.
3	27.543763004	PCSSystemtec_44:dd:...	PCSSystemtec_44:dd:...	ARP	60	192.168.56.102 is at 08:00:27:64:8f:7d
4	27.543896731	PCSSystemtec_04:8f:...	PCSSystemtec_44:dd:...	ARP	60	Who has 192.168.56.101? Tell 192.168.
2008	45.632861624	PCSSystemtec_04:8f:...	PCSSystemtec_44:dd:...	ARP	60	Who has 192.168.56.101? Tell 192.168.
2009	45.632862563	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.101 is at 08:00:27:64:8f:7d
2829	146.121721487	PCSSystemtec_04:8f:...	PCSSystemtec_9f:7d:...	ARP	60	Who has 192.168.56.100? Tell 192.168.
2830	146.121722097	PCSSystemtec_04:8f:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.100 is at 08:00:27:64:8f:7d
2973	212.895695735	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2974	214.896258738	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2975	216.897210852	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2976	218.900000000	PCSSystemtec_04:8f:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2977	220.898175293	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2978	222.899173208	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2979	224.900000004	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2980	226.900666629	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2981	228.902330719	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2982	230.903079421	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2983	232.903592256	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2984	233.810299906	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d
2985	234.810794726	PCSSystemtec_44:dd:...	PCSSystemtec_04:8f:...	ARP	60	192.168.56.103 is at 08:00:27:64:8f:7d

ARP spoofing packets capture



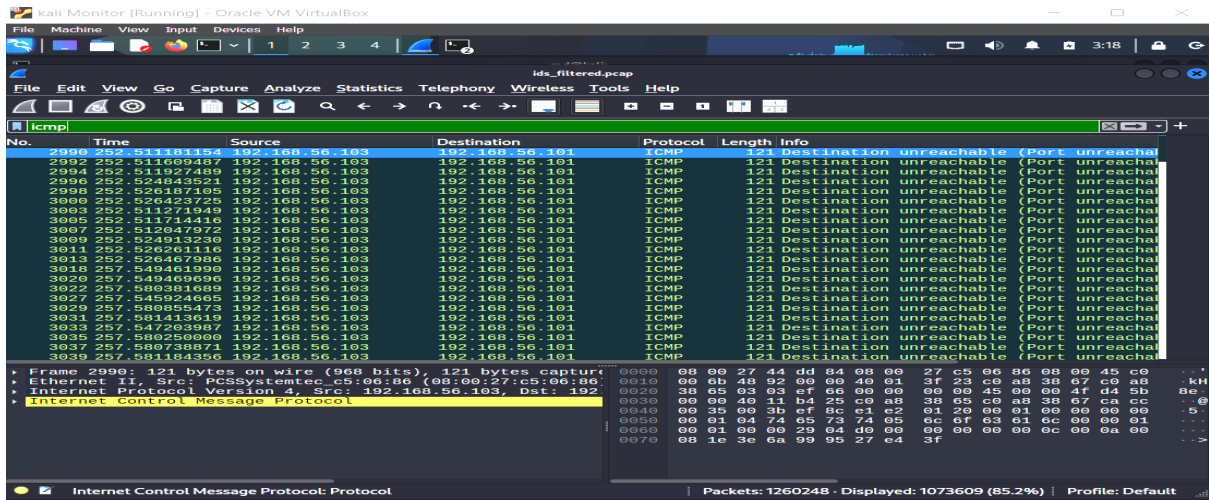
The screenshot shows a Wireshark capture of DNS traffic. The packet list on the left shows several DNS queries from source 192.168.56.101 to destination 192.168.56.103. The selected packet (No. 2990) is a Standard query for 0xe1e2 A test.local OPT. The packet details pane shows the query structure, and the packet bytes pane shows the raw data. The status bar at the bottom indicates 1260248 packets displayed, with 102 (0.0%) shown.

No.	Time	Source	Destination	Protocol	Length	Info
2980	252.511139972	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xe1e2 A test.local OPT
2990	252.511181154	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2991	252.511592767	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xe1e2 A test.local OPT
2992	252.511609487	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2993	252.511921993	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xe1e2 A test.local OPT
2994	252.511927489	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2995	252.524795563	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xcbb4 A test.local OPT
2996	252.524812191	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2997	252.526169581	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xcbb4 A test.local OPT
2998	252.526187165	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2999	252.526431319	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xe1e2 A test.local OPT
3000	252.526423725	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3001	252.524241894	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xb1b3 A test.local OPT
3002	252.51115038	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xe1e2 A test.local OPT
3003	252.511271949	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3004	252.511598984	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xe1e2 A test.local OPT
3005	252.511741416	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3006	252.511929649	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xe1e2 A test.local OPT
3007	252.512047072	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3008	252.524912902	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xcbb4 A test.local OPT
3009	252.524913230	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3010	252.526431319	192.168.56.101	192.168.56.103	DNS	93	Standard query 0xcbb4 A test.local OPT

Domain Name System: Protocol

Packets: 1260248 · Displayed: 102 (0.0%) · Profile: Default

DNS query packets capture



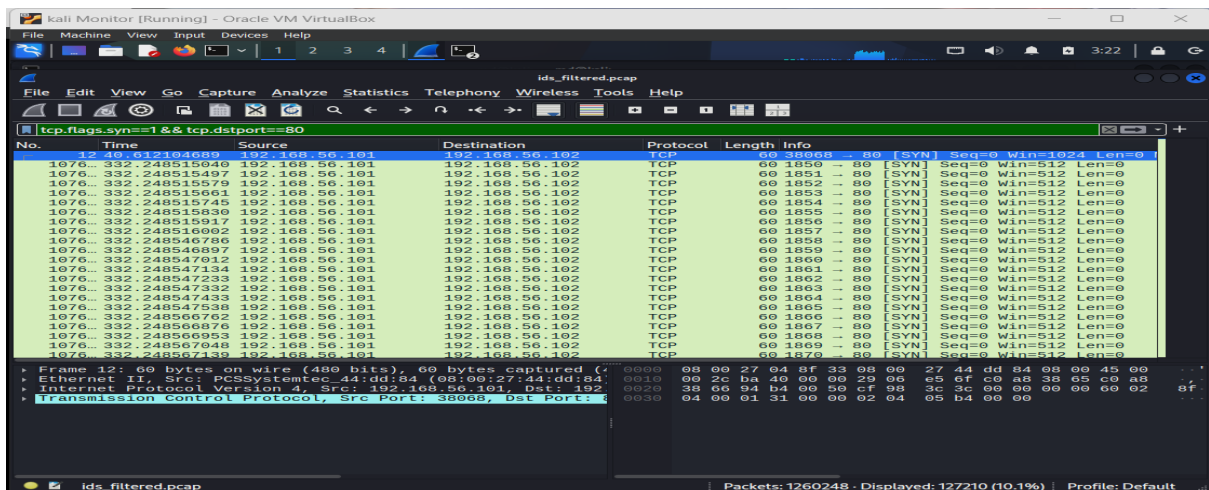
The screenshot shows a Wireshark capture of ICMP traffic. The packet list on the left shows several ICMP destination unreachable messages from source 192.168.56.103 to destination 192.168.56.101. The selected packet (No. 2990) is an ICMP destination unreachable message. The packet details pane shows the ICMP structure, and the packet bytes pane shows the raw data. The status bar at the bottom indicates 1260248 packets displayed, with 1073609 (85.2%) shown.

No.	Time	Source	Destination	Protocol	Length	Info
2990	252.511181154	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2991	252.511609487	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2992	252.511927489	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2993	252.524843521	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2994	252.526187195	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2995	252.526423725	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2996	252.511271949	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2997	252.511741416	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2998	252.512047072	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
2999	252.524912902	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3000	252.526261116	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3001	252.526467986	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3002	257.549619990	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3003	257.549646969	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3004	257.549924665	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3005	257.580855473	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3006	257.581343619	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3007	257.584279387	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3008	257.580858000	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3009	257.580738871	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)
3010	257.581184356	192.168.56.103	192.168.56.101	ICMP	121	Destination unreachable (Port unreachable)

Internet Control Message Protocol: Protocol

Packets: 1260248 · Displayed: 1073609 (85.2%) · Profile: Default

ICMP packets capture



The screenshot shows a Wireshark capture of SYN flood traffic. The packet list on the left shows several TCP SYN packets from source 192.168.56.101 to destination 192.168.56.102. The selected packet (No. 1299) is a TCP SYN packet. The packet details pane shows the TCP structure, and the packet bytes pane shows the raw data. The status bar at the bottom indicates 1260248 packets displayed, with 127210 (10.1%) shown.

No.	Time	Source	Destination	Protocol	Length	Info
1299	332.248510469	192.168.56.101	192.168.56.102	TCP	60	1850 → 80 [SYN] Seq=0 Win=1024 Len=0
1299	332.248515940	192.168.56.101	192.168.56.102	TCP	60	1850 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248515579	192.168.56.101	192.168.56.102	TCP	60	1852 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248515661	192.168.56.101	192.168.56.102	TCP	60	1853 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248515745	192.168.56.101	192.168.56.102	TCP	60	1854 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248515830	192.168.56.101	192.168.56.102	TCP	60	1855 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248515917	192.168.56.101	192.168.56.102	TCP	60	1856 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248516002	192.168.56.101	192.168.56.102	TCP	60	1857 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248546786	192.168.56.101	192.168.56.102	TCP	60	1858 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248546897	192.168.56.101	192.168.56.102	TCP	60	1859 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248547612	192.168.56.101	192.168.56.102	TCP	60	1860 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248547134	192.168.56.101	192.168.56.102	TCP	60	1861 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248547233	192.168.56.101	192.168.56.102	TCP	60	1862 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248547332	192.168.56.101	192.168.56.102	TCP	60	1863 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248547433	192.168.56.101	192.168.56.102	TCP	60	1864 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248547538	192.168.56.101	192.168.56.102	TCP	60	1865 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248560762	192.168.56.101	192.168.56.102	TCP	60	1866 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248560876	192.168.56.101	192.168.56.102	TCP	60	1867 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248560953	192.168.56.101	192.168.56.102	TCP	60	1868 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248567048	192.168.56.101	192.168.56.102	TCP	60	1869 → 80 [SYN] Seq=0 Win=512 Len=0
1299	332.248567139	192.168.56.101	192.168.56.102	TCP	60	1870 → 80 [SYN] Seq=0 Win=512 Len=0

ids_filtered.pcap

Packets: 1260248 · Displayed: 127210 (10.1%) · Profile: Default

SYN flood packets capture

Remediation Recommend:

Immediate actions for critical severity :

Disabling SSH password authentication would be the right choice and enable public key authentication exclusively, the more encrypted it is the more secure it will be.

Change SSH from default port 22 to non standard port that can be helped in reduction of any automated surface level attack. All services should be enforced with strong password policies and lockout of accounts.

Short term actions for high severity:

Enabling the SYN cookies on servers while limiting the TCP connection rate would be the right choice as all the cookies will mitigate the DoS attacks. To prevent ARP spoofing on all managed switches the Dynamic ARP Inspection should be enabled. Deploy network segmentation to limit lateral movement potential in the event of compromise to prevent the access control of any unauthorized devices. On the other hand , to prevent volumetric ping floods ICMP rate limit on perimeter firewall should be configured. Similarly implementing DNS query rate limit and anomaly detection will prevent any DNS tunneling exfiltration.

Long term actions for high severity:

The SIEM integration is required to be extended to more log sources (system logs, application logs, firewall logs) and the configure automated alerts in Splunk to alert the SOC team when critical thresholds are exceeded. On regular penetration testing the emerging vulnerabilities can be detected and prevent any further exploitation. The SIEM alert correlation should be developed as a formal incident response playbook.

Conclusion:

This project successfully delivered a full SIEM integrated Network Intrusion Detection System with SOC platform built from scratch in a controlled Virtual lab. Six distinct attack vectors were generated and detected in real time where total attacks were 21, among them 2 were in critical alerts and the rest 19 were on high severity, with all the alerts automatically forwarded to splunk enterprise for log analysis.

The project demonstrates full SOC workflow competency covering network traffic capture and real-time threat detection, automated log ingestion and visualization on a SIEM dashboard, and formal vulnerability reporting with CVSS scoring and remediation guidance. This is similar to the main tasks of a real world Security Operations Centre analyst.

The severity ratings of all six are HIGH or CRITICAL (CVSS 7.5-9.8) underscoring the critical importance of layered network security controls.

Appendix:

- [Scapy Documentation](#)
- [Splunk Enterprise Documentation](#)
- [NIST CVSS v3.1 Calculator](#)
- [VulnHub — Metasploitable3](#)